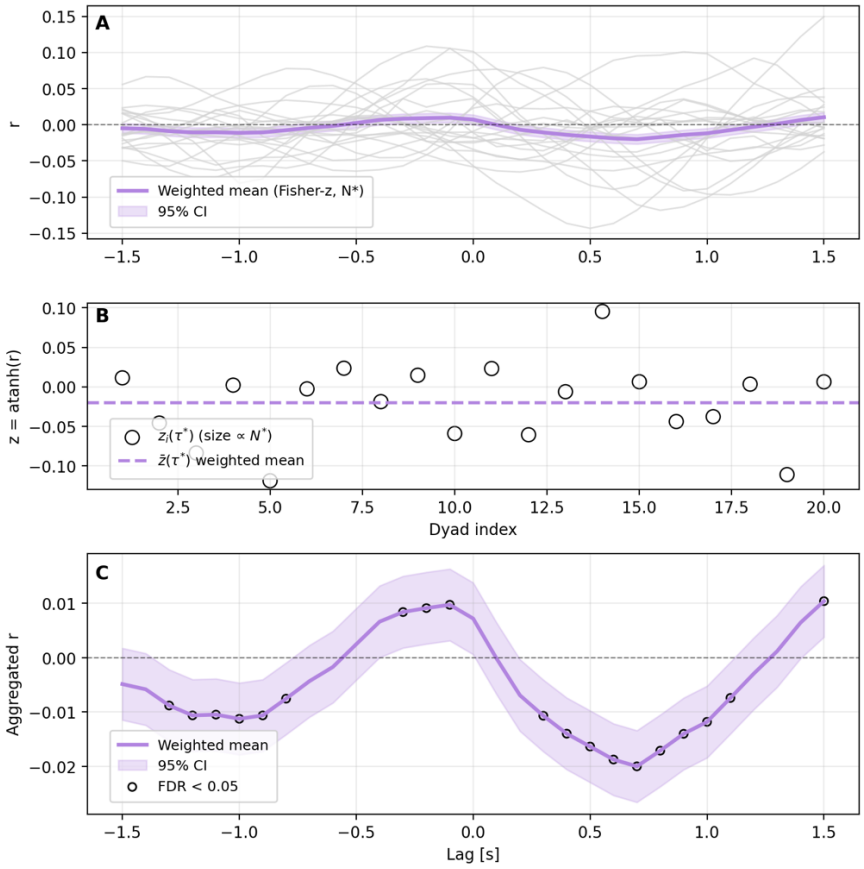
Supplementary Material 1 – Aggregation

Aggregation of cross-correlation curves

This section complements the description provided in the manuscript. Each pair of participants ("dyad") produces its own cross-correlation curve, showing how their movements covary across different time lags (for example, from –1.5 s to +1.5 s). Each point on these curves represents how strongly the two individuals are coordinated at that specific temporal offset.

To obtain one representative curve per condition, all dyadic curves are combined into a single group-level estimate. Because correlation coefficients (r) are not additive, we do not use a simple arithmetic mean. Instead, each ri(τ) is first transformed using the Fisher-z transformation, which linearizes the correlation scale and allows proper statistical averaging.

The transformed values are then averaged across dyads using effective sample size weights (N∗), and finally converted back to the r scale. This Fisher-z weighted approach is standard in meta-analysis and neuroimaging when combining correlations across subjects or sessions.



Illustrative pipeline (A–C).

The above figure illustrates this process in three steps:
A (Individual dyads): Each gray line shows one dyad's correlation curve $r_i(\tau)$ across time lags. The colored line represents the weighted mean curve, and the shaded band its 95 % confidence interval. This panel displays both inter-dyad variability and the overall group trend.

B (Averaging step at a specific lag): This panel isolates a single time lag, denoted τ^*, corresponding to the point of maximum overall coordination. Each dot represents the Fisher-z transformed value $z_i(\tau^*)=\operatorname{atanh}(r_i(\tau^*))$ for a specific dyad.
The x-axis represents the dyad index (Dyad 1, Dyad 2, Dyad 3, …, n), meaning each position corresponds to a different pair of participants — not to time.
The y-axis represents the transformed correlation value $z_i(\tau^*)$ for that dyad.
The dashed horizontal line marks the weighted mean $z(\tau^*)$ computed as:

$$\bar{z}(\tau^*) = \frac{\sum_i N_i^* \, z_i(\tau^*)}{\sum_i N_i^*}$$

This line represents the group-level average for that specific lag. The same averaging process is then repeated across all lags to build the final group-level curve. Panel B is key to understanding the aggregation process. It shows that each point on the group curve results from combining the individual dyadic correlations at the same time lag through a weighted mean in Fisher-z space.

C (Aggregated curve): The final panel displays the aggregated correlation curve after converting back from the Fisher-z to the r scale. The colored line represents the group mean, the shaded area shows its 95 % confidence interval, and open markers highlight time lags that remain significant after FDR correction.